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Bayesian Neural Surrogates Unify Uncertainty Propagation, Sensitivity Attribution and Calibration for Coupled Multi-Physics Reactor Analysis

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Abstract

We train a Bayesian neural network (BNN) surrogate on coupled OpenMC–FEniCS data of a heat-pipe-cooled microreactor and use its posterior predictive distribution as the single probabilistic object on which forward uncertainty propagation, Sobol sensitivity attribution and MCMC posterior calibration all operate, without retraining or recalibration between analyses. Two-way neutronic–thermal–structural coupling damps stress variability by approximately 28% relative to a decoupled prediction; stress and reactivity are governed by largely separate material parameters ($E_{\text{intercept}}$ and α_{base}, respectively); and across 18 benchmark calibrations the posterior contracts toward observation-compatible regions with 90%-CI coverage of 0.917. Each posterior predictive draw costs approximately 1.76×10^5 times less than a coupled high-fidelity solve, enabling the full uncertainty workflow to run end-to-end on a single workstation. The posterior-predictive framework is directly applicable to other coupled multi-physics systems.

1 Introduction

Microreactors—compact, transportable fission systems rated at 1–20 $\mathrm{MW_e}$—are being developed for decentralized power generation in remote communities, defense installations and space missions[1,11]. NASA and DOE are advancing fission surface-power systems for sustained lunar and planetary operations[12], while rapidly growing electricity demand from AI-intensive data centres reinforces the case for distributed, dispatchable zero-carbon generation[13]. Unlike conventional large-scale plants, these designs concentrate fission power, heat removal and structural loading within a compact monolithic core where neutronics, heat transfer and structural mechanics are tightly coupled through two-way feedback[1,2]. A single uncertain material parameter can simul-

taneously alter the neutronic balance, the thermal field and the stress state; the closed feedback loop may amplify or damp the resulting variability in ways that single-physics analyses cannot capture. The uncertainty problem in these systems is therefore inherently coupled.

High-fidelity coupled solvers now exist for these systems: MOOSE-based toolchains handle neutronics–thermo-mechanics–heat-pipe coupling for KRUSTY-class and larger designs[2,3], and thermal–structural analyses have established stress concentration as a persistent safety concern in monolithic-core configurations[4,5]. Surrogate-assisted probabilistic modules in the MOOSE ecosystem[6,7] and digital-twin pipelines[8] address forward propagation, sensitivity or calibration individually, but each uses a separately constructed model with no shared probabilistic basis between analyses.

Running the full probabilistic workflow—forward uncertainty propagation, global sensitivity analysis and Bayesian calibration—directly on the coupled solver is impractical at $\sim 10^3$ s per solve. Gaussian-process and polynomial-chaos surrogates face dimensionality constraints in nonlinear coupled settings; physics-informed neural networks require a closed residual form that iterative solver exchange does not supply; deterministic neural networks produce only point predictions. None of these provides the coherent predictive distribution that all three analyses require.

A Bayesian neural network (BNN) fills this gap. By placing a distribution over the network weights, a BNN yields a posterior predictive distribution that can be sampled at negligible cost once trained. This single distribution serves as the shared probabilistic layer on which forward propagation, variance-based sensitivity attribution and observation-conditioned posterior calibration all operate, with no retraining or recalibration between them.

This paper presents a BNN-based posterior-predictive framework for end-to-end probabilistic analysis of coupled multi-physics systems. The framework unifies forward uncertainty propagation, Sobol sensitivity decomposition and MCMC posterior calibration on a single predictive layer trained on coupled OpenMC–FEniCS data of a heat-pipe-cooled reactor. It reveals that two-way coupling damps stress variability by approximately 28%, that stress and reactivity are governed by largely separate parameter pathways, and that the posterior calibration contracts the material-parameter distribution toward observation-compatible regions—all at five orders of magnitude less cost per evaluation than a coupled high-fidelity solve.

2 Results

2.1 Surrogate accuracy and calibration

The framework rests on a single BNN that predicts all 15 coupled and decoupled outputs simultaneously; if this surrogate is inaccurate, the downstream propagation, sensitivity and calibration results are unreliable. We evaluated the physics-regularized BNN alongside a reference (unconstrained) BNN and two conventional baselines—MC-Dropout and a five-member deep ensemble—on 435 held-out test samples (Table 1, Fig. 1). Results for the physics-regularized BNN are reported as mean ± s.d. across 5 random seeds; ablation across four BNN variants is in Supplementary Table S3.

On coupled peak global stress the physics-regularized BNN achieves $R^2 = 0.935 \pm 0.007$ (RMSE 7.63 ± 0.47 MPa), comparable to MC-Dropout ($R^2 = 0.934$) and deep ensemble ($R^2 = 0.934$).

The BNN advantage is most pronounced on temperature outputs: for maximum fuel temperature the BNN RMSE is 4.26 ± 0.18 K versus 4.50 K for both baselines, and the expected calibration error (ECE) is 0.027 versus 0.183 (MC-Dropout) and 0.190 (deep ensemble)—a five- to six-fold improvement in calibration quality (Fig. 1A, C).

The 90% predictive intervals are conservatively calibrated across all outputs (PICP 0.937–1.000), and incorporating physics-prior monotonicity constraints tightens the intervals relative to the unconstrained reference without reducing coverage (Supplementary Table S3). The continuous ranked probability score (CRPS) is lowest for the physics-regularized BNN on four of five primary outputs (Table 1).

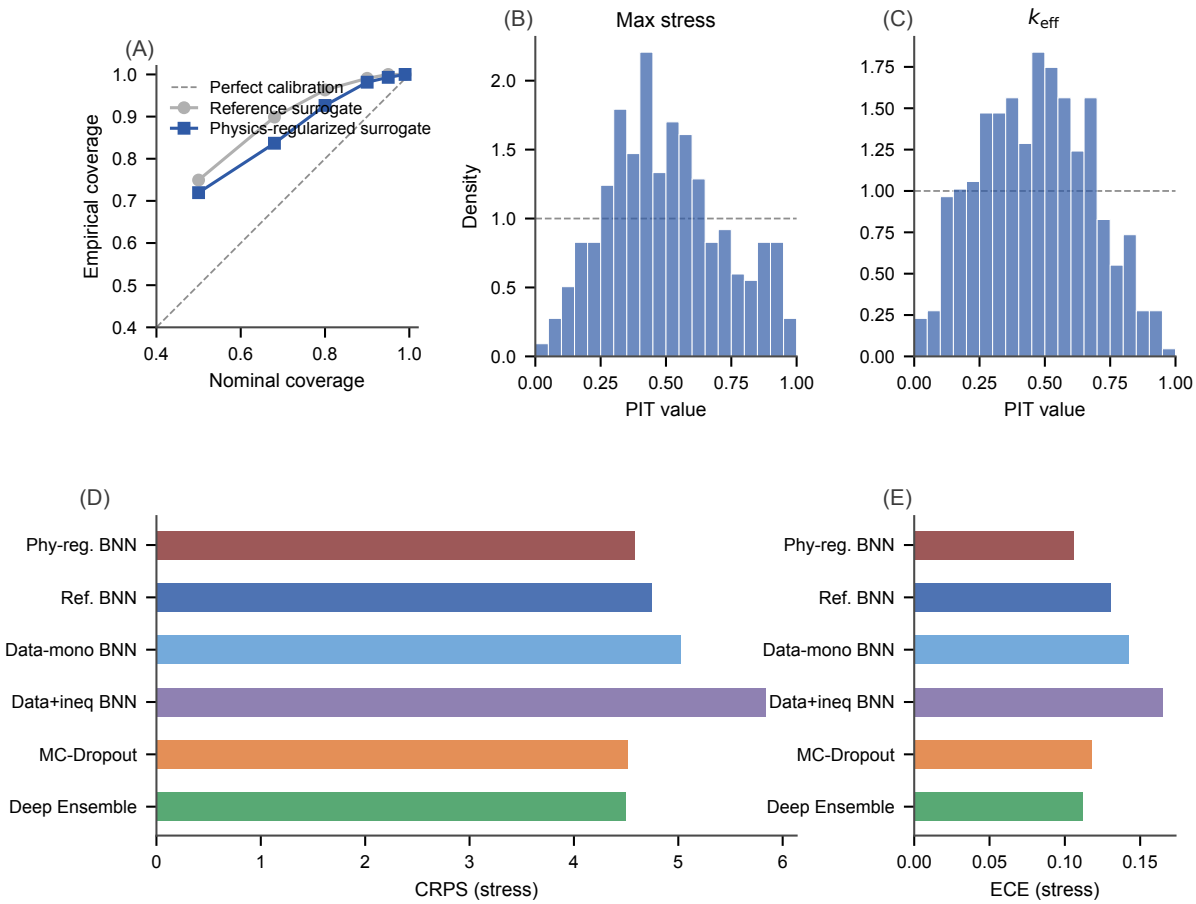

Figure 1: **Surrogate accuracy and uncertainty calibration.** (**A**) Predicted vs. true values for five primary coupled outputs (physics-regularized BNN, 435 test samples). (**B**) Residual distributions. (**C**) Calibration curves comparing BNN, MC-Dropout and deep ensemble. The BNN achieves five- to six-fold lower ECE than the baselines on temperature outputs.

2.2 Physics consistency

We verified the BNN's adherence to known physical relationships through two tests: monotonicity verification against 15 input–output pairs derived from the constitutive laws, and inequality constraint satisfaction (Fig. 2A). For all high-confidence monotonicity pairs—including the dominant stress drivers $E_{\mathrm{intercept}}$ (+), α_{base} (+), and k_{ref} (−)—all four BNN variants achieved zero violation rates on the 435 held-out test samples using discrete $\pm\delta$ perturbation testing. The only

Table 1: **Surrogate accuracy on five primary coupled outputs.** Physics-regularized BNN: mean ± s.d. across 5 seeds; external baselines: single run. Bold CRPS indicates best per output.

Output	R^2	RMSE	PICP	ECE	CRPS
Physics-regularized BNN					
k_{eff}	0.633 ± 0.258	0.0006 ± 0.0004	0.981 ± 0.007	0.123	**0.0002**
Max. fuel temp.	0.621 ± 0.044	4.26 ± 0.18 K	0.936 ± 0.015	0.027	**2.39**
Max. monolith temp.	0.622 ± 0.044	4.80 ± 0.20 K	0.942 ± 0.013	0.031	**2.70**
Max. global stress	0.935 ± 0.007	7.63 ± 0.47 MPa	0.987 ± 0.007	0.106	**4.58**
Wall expansion	0.993 ± 0.001	0.0020 ± 0.0002	1.000 ± 0.000	0.194	0.0017
MC-Dropout					
Max. fuel temp.	0.590	4.50 K	—	0.183	2.62
Max. global stress	0.934	7.66 MPa	—	0.118	4.52
Deep Ensemble (5 members)					
Max. fuel temp.	0.590	4.49 K	—	0.190	2.63
Max. global stress	0.934	7.64 MPa	—	0.112	4.50

non-trivial violations occurred for the medium-confidence pair SS316_α → fuel temperature (36–54% across models), reflecting a weaker and potentially nonlinear relationship that the training data does not consistently support. All four models also satisfied ordering constraints (max ≥ average temperature) and non-negativity constraints (stress ≥ 0) with zero violations.

Decomposing total predictive variance into epistemic (weight-uncertainty) and aleatoric (observation-noise) components revealed that aleatoric uncertainty dominates across all outputs, accounting for 58–90% of total variance (Fig. 2B). The epistemic fraction was highest for wall expansion (∼42%) and stress (∼30%), indicating greater model-form uncertainty for these mechanically driven outputs.

2.3 Forward uncertainty propagation

Propagating 20,000 Latin hypercube samples through the BNN posterior predictive mean yields a coupled stress distribution with mean 162.5 MPa and standard deviation 35.5 MPa, versus a decoupled mean of 202.4 MPa and standard deviation of 49.2 MPa. Coupling reduces the stress standard deviation by approximately 28% and lowers the mean by approximately 40 MPa (Table 2, Fig. 3).

The coupled k_{eff} has mean 1.1036 and standard deviation approximately 107 pcm. The decoupled k_{eff} is effectively constant because, at the reference geometry, the sampled material parameters do not enter the neutronic cross-section tables; all coupled k_{eff} variability originates from thermal-expansion modification of core geometry.

Table 2: **Forward UQ summary: coupled vs. decoupled distributions** (20,000 LHS samples through BNN posterior predictive mean).

Output	Coupled mean	Coupled s.d.	Decoupled mean	Decoupled s.d.
Stress (MPa)	162.5	35.5	202.4	49.2
k_{eff}	1.1036	0.00107	∼constant	∼0

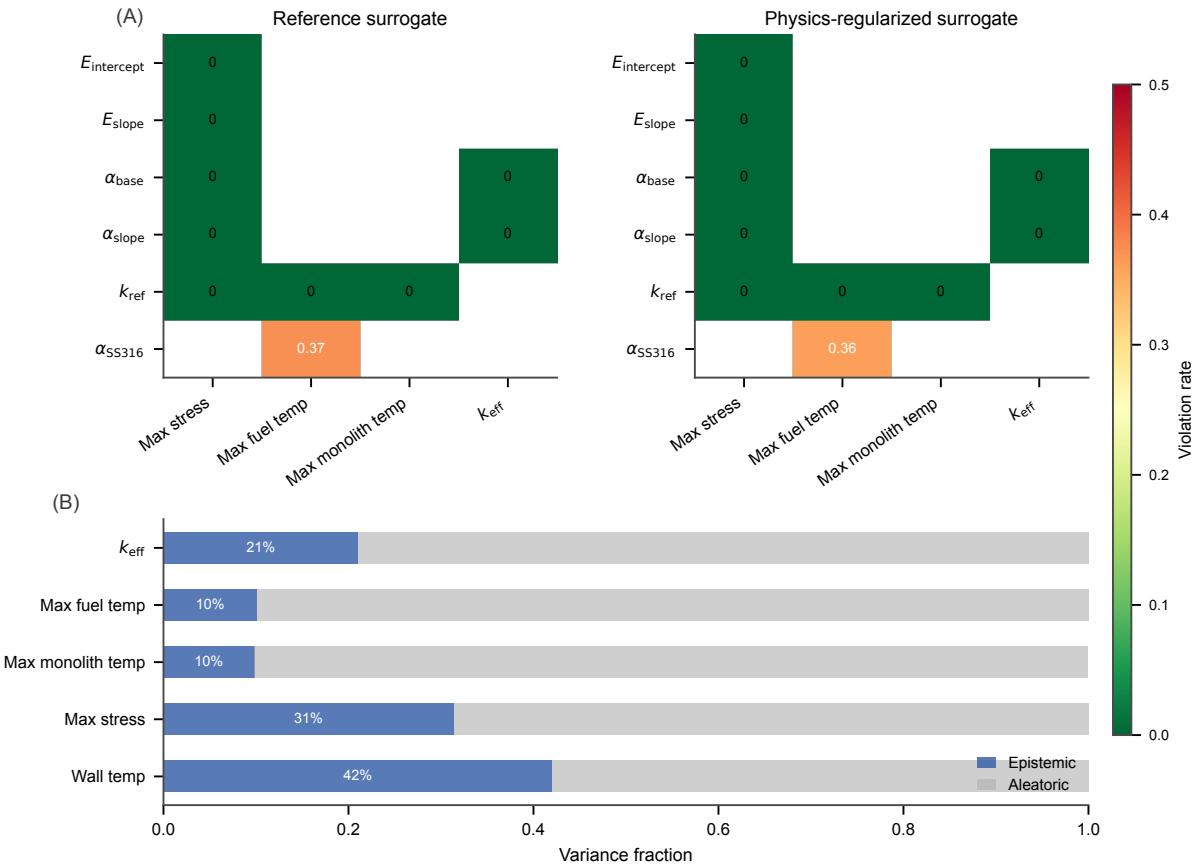

Figure 2: **Physics consistency verification.** (**A**) Monotonicity violation rates for 15 input–output pairs across four BNN variants; high-confidence pairs achieve zero violations. (**B**) Epistemic vs. aleatoric variance decomposition across five primary outputs.

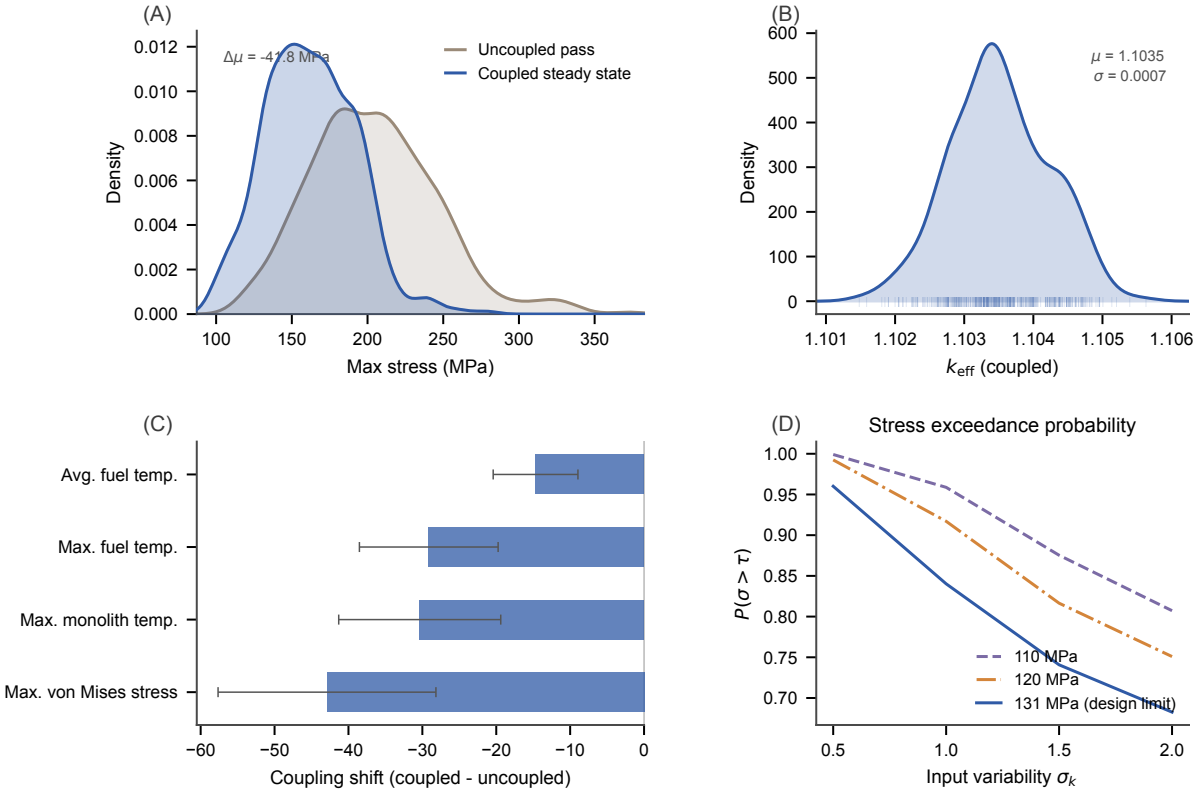

Figure 3: **Forward uncertainty propagation.** Coupled vs. decoupled output distributions for stress and k_{eff}. Two-way coupling reduces the stress standard deviation by ≈28% and introduces k_{eff} variability that is absent in the decoupled case.

2.4 Global sensitivity analysis

Sobol analysis (50 independent replications, $N_S = 4096$ per matrix; full index table in Supplementary Table S4) shows that stress and k_{eff} are governed by two largely separate physical pathways (Table 3, Fig. 4).

Stress is rooted in the elastic-constitutive pathway. $E_{\text{intercept}}$ alone accounts for approximately 55% of the first-order variance ($S_1 = 0.548$, 90% CI [0.498, 0.572]), and the near-zero gap between its total-order and first-order indices indicates that it acts with minimal interaction. Secondary contributors—α_{base} ($S_1 = 0.210$), k_{ref} and ν—add a collective approximately 15%, while the remaining parameters are negligible. The dominance of $E_{\text{intercept}}$ is a direct consequence of the generalised Hooke's law: stiffness enters the stress field linearly.

k_{eff} is rooted in the thermal-expansion pathway. α_{base} accounts for approximately 77% of the first-order variance ($S_1 = 0.771$, 90% CI [0.747, 0.788]): it sets the magnitude of geometric expansion, which alters neutron leakage and shifts the neutronic balance. α_{slope} contributes a further approximately 19% ($S_1 = 0.195$), while no elasticity or conductivity parameter has a first-order effect on reactivity above the noise floor.

The two pathways share almost no dominant parameters. This separation means that stress and k_{eff} observations constrain different regions of the material-parameter space—an important constraint for any monitoring strategy that aims to reduce material-property uncertainty.

Table 3: **First-order Sobol indices S_1 for stress and k_{eff}** (physics-regularized BNN, $N_S = 4096$, 50 replications, 90% CI). Only factors with CI lower bound > 0 are shown.

Parameter	Stress S_1 [90% CI]	k_{eff} S_1 [90% CI]
$E_{\text{intercept}}$	0.548 [0.498, 0.572]	—
α_{base}	0.210 [0.171, 0.254]	0.771 [0.747, 0.788]
ν	0.063 [0.034, 0.097]	—
k_{ref}	0.065 [0.014, 0.118]	—
α_{slope}	—	0.195 [0.112, 0.250]

2.5 Posterior calibration

Bayesian calibration uses the same BNN to update the prior parameter distribution given noisy observations. The four calibrated parameters—$E_{\text{intercept}}$, α_{base}, α_{slope} and k_{ref}—are the four that the Sobol analysis identified as covering the dominant variance channels for stress and k_{eff}. Metropolis–Hastings MCMC is run on 18 benchmark cases drawn from the test split (6 low / 6 near / 6 high stress; details in Methods) with 4 independent chains per case.

Chain convergence was assessed using rank-normalised split-$\hat{R}$ (Vehtari et al. 2021) and Geyer initial-positive-sequence effective sample size. Across all 72 parameter–case combinations for the physics-regularized BNN, $\hat{R}_{\max} = 1.017$ (well below the 1.1 threshold), $\text{ESS}_{\min} = 352$, and mean acceptance rate = 0.605 (Fig. 5A–B; Supplementary Figs. S9–S10 and Supplementary Table S5).

Across the 18 cases, the posterior 90%-CI coverage is 0.917—the observations successfully constrain the parameters that the Sobol analysis flagged as most influential. In higher-stress cases,

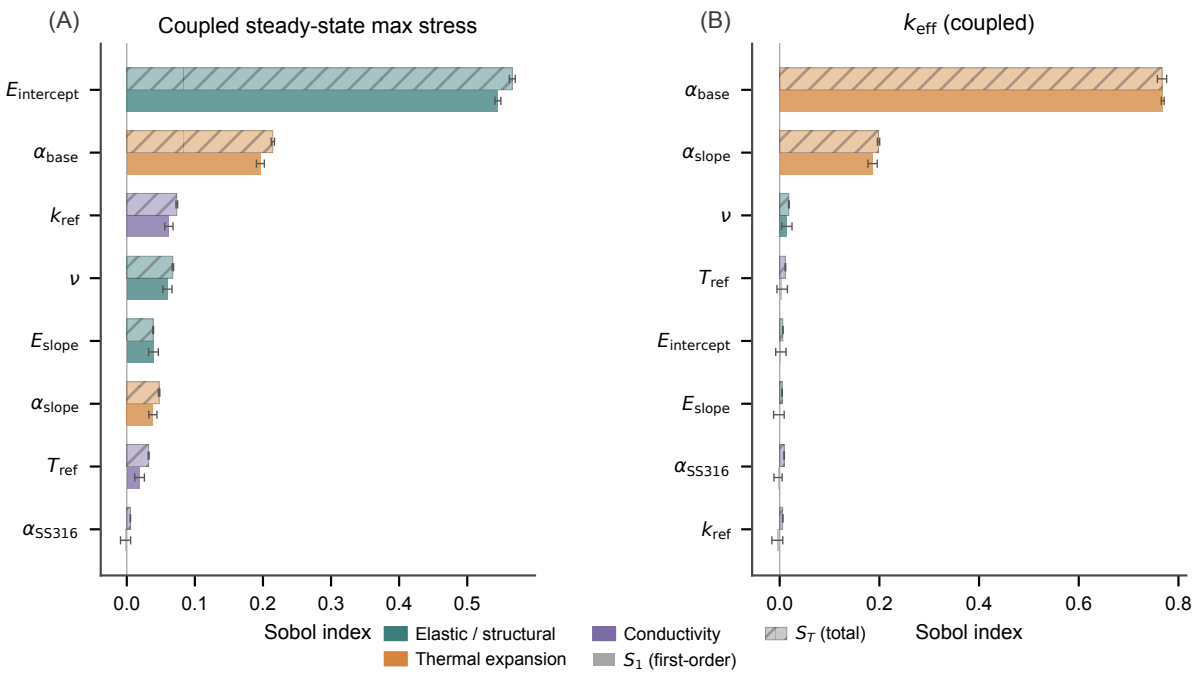

Figure 4: **Sobol sensitivity analysis.** First-order and total-order indices for stress and k_{eff} reveal two largely separate physical pathways: the elastic-constitutive pathway (dominated by $E_{\text{intercept}}$) for stress, and the thermal-expansion pathway (dominated by α_{base}) for k_{eff}.

the $E_{\text{intercept}}$ marginal shifts toward larger values, consistent with its dominant role in the elastic-constitutive pathway. The joint posterior of $E_{\text{intercept}}$ and α_{base} develops a compensation ridge: an increase in stiffness can be partially offset by a decrease in thermal expansion, leaving the stress product $\sigma \sim E \cdot \alpha \cdot \Delta T$ approximately invariant. This partial non-identifiability arises from the multiplicative structure and would not be visible in a single-output calibration (Fig. 5).

At the posterior-mean parameters of each case, the BNN-predicted stress tracks the true value with MAE = 10.4 MPa over a true-stress range of 102–198 MPa. Prior sensitivity analysis (6 prior configurations) and observation noise sensitivity analysis (noise fraction 0.5–10%) confirmed the robustness of posterior inference (Supplementary Figs. S14–S15, Tables S11–S12).

2.6 Computational efficiency

The BNN produces one posterior-predictive draw (averaged over 50 Monte Carlo weight samples) in 1.29×10^{-2} s on GPU. Batched inference at batch size 1024 reaches a per-sample cost of 1.34×10^{-5} s, equivalent to a throughput of approximately 7.5×10^4 samples/s. The coupled HF solver requires 2266 s (mean of three independent runs, s.d. 1.34 s; AMD EPYC 9654 96-core node, 566 GB RAM, 2× RTX 5090). The resulting speedup is approximately 1.76×10^5 for one posterior-predictive draw and approximately 1.69×10^8 for batched per-sample inference. A 20,000-sample forward-UQ pass requires approximately 0.3 s of GPU time, against 4.5×10^7 s of sequential HF cost. This speedup enables the full probabilistic workflow—20,000-sample Monte Carlo propagation, 50-replication Sobol analysis, and 18-case MCMC calibration—to run within minutes of wall-clock time.

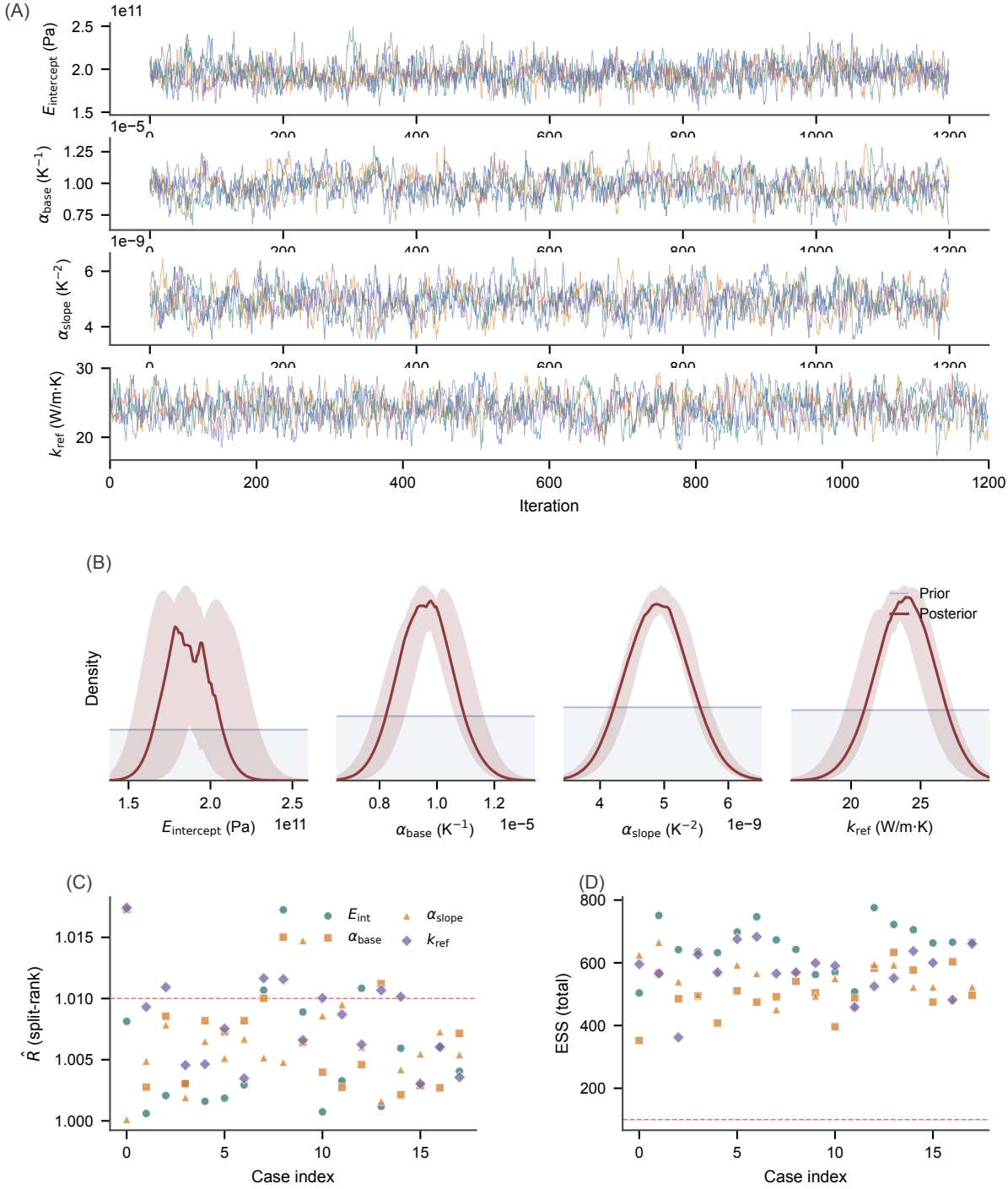

Figure 5: **Posterior calibration results.** (**A**) MCMC convergence diagnostics (split-$\hat{R}$ and ESS). (**B**) Acceptance rates across 18 benchmark cases. (**C**) Marginal posterior distributions for $E_{\text{intercept}}$ and α_{base} in representative low-, near-, and high-stress cases. (**D**) Joint posterior of $E_{\text{intercept}}$ vs. α_{base} showing the compensation ridge.

The BNN also exploits limited training data efficiently. With only 507 training samples (25% of the full training set), the surrogate achieves RMSE = 4.93—96% of its peak performance at full data utilisation (RMSE = 4.62). Marginal improvement from 50% to 100% data is less than 1% (Fig. 6).

3 Discussion

The stress standard deviation drops from approximately 49.2 MPa in the decoupled response to approximately 35.5 MPa after coupling convergence, a roughly 28% reduction. The mechanism is the negative neutronic–thermal feedback characteristic of compact monolithic-core systems: thermal expansion modifies the core geometry, which shifts the neutronic balance and partially offsets the direct stress increase. What the BNN-based forward propagation establishes is the net effect of this feedback on the output distribution. A decoupled analysis does not see the feedback loop and therefore overestimates the parameter-driven stress variability. The overestimate is conservative for safety-margin evaluation and cannot produce an unsafe conclusion, but it inflates the apparent penalty of material-property uncertainty. In practice, this can translate into manufacturing tolerances stricter than the coupled response warrants.

Compared with an ELBO-only BNN (Supplementary Table S3), adding physics-prior monotonicity constraints shrinks the 90% predictive interval without reducing coverage. The primary model enforces monotonicity relations derived from the physics of the constitutive laws—for example, that stress increases with Young's modulus at fixed thermal strain. The interval tightening at constant coverage indicates genuine sharpening rather than miscalibration, and the same posterior predictive distribution propagates consistently through forward UQ, Sobol attribution and MCMC calibration. A point-estimate surrogate would require separate uncertainty wrappers for each analysis with no guarantee of mutual consistency.

The sensitivity and calibration results have practical consequences for the engineering workflow. For stress-margin assessment, tightening the uncertainty on $E_{\text{intercept}}$—e.g. through coupon tensile tests at operating temperature—is the single highest-leverage action; for reactivity stability, dilatometry to refine α_{base} is the priority. Because these two measurements target different experimental setups, they can proceed in parallel. When measured data become available, the posterior calibration folds that information into the same surrogate without retraining. If a posterior update reveals that a safety margin is tighter than initially expected, the Sobol ranking identifies which parameter to prioritise in the next round of characterisation. The near-zero parameter overlap between the stress and k_{eff} pathways means that a calibration protocol combining both outputs provides stronger parameter identification than either alone. These indications are model-driven and require verification on the high-fidelity solver before adoption as design directives.

The surrogate is trained on data from a single HPR design point; a different geometry, fuel enrichment or operating envelope would require regenerating the training set and retraining. The methodology itself is not design-specific: the same workflow of coupled-data generation, BNN training and three-stage probabilistic analysis is directly applicable to other reactor types and other coupled multi-physics systems. The MCMC calibration fixes four parameters at their true values and calibrates the remaining four; extending to a full eight-parameter calibration would

require additional independent observations. The k_{eff} surrogate degrades on out-of-distribution α_{base} tails (R^2 drops to approximately 0.29 versus 0.825 in-distribution; Supplementary Fig. S5), so the Sobol and posterior conclusions for k_{eff} hold only within the training range. All posterior summaries are computed on the surrogate side. An independent high-fidelity re-evaluation at the posterior-mean parameter vectors is documented separately. The present study uses synthetic observations constructed from the high-fidelity solver; if experimental or monitoring data become available, the same calibration workflow can ingest real observations directly.

4 Methods

4.1 Material uncertainty model

The coupled HPR simulation involves eight material properties of the SS316 monolith grouped by physical function: structural mechanics ($E_{\text{intercept}}$, E_{slope}, ν), thermal deformation (α_{base}, α_{slope}), and thermal transport (k_{ref}, T_{ref}, k_{slope}). All eight are assigned independent uniform priors with half-width 10% of the nominal design value. The same prior distribution is used throughout: as the sampling distribution in forward UQ, as the input domain for Sobol analysis, and as the prior in posterior calibration. Full parameter definitions, nominal values and constitutive-law relationships are in Supplementary Note E and Supplementary Table S1.

4.2 Coupled high-fidelity simulation

The high-fidelity workflow couples OpenMC Monte Carlo neutronics with the FEniCS finite-element solver through a Picard fixed-point iteration scheme[9]. Two fidelity levels are retained: a decoupled single-pass prediction before coupling convergence and a coupled steady-state prediction after full convergence. Each HF evaluation produces 15 scalar outputs. A dataset of $n = 2900$ samples was generated by Latin hypercube sampling; samples were split into training (2029), validation (436) and test (435) sets with a fixed random seed (seed = 2026). The BNN maps directly from the 8 input parameters to the 15 outputs without resolving intermediate coupling fields. Details of the iteration scheme, convergence criteria and output definitions are in Supplementary Note E.

4.3 Bayesian neural network surrogate

The surrogate is a BNN with mean-field variational inference (MFVI, Bayes-by-Backprop[10]). The network uses a fully connected backbone with two output heads: a mean head and a log-variance head that models heteroscedastic observation noise. Training minimises the negative evidence lower bound (ELBO). The posterior predictive distribution at a new input is approximated by Monte Carlo averaging over M weight samples from the trained variational posterior: at inference $M = 50$, for Sobol analysis $M = 30$, and for the MCMC likelihood $M = 20$. Hyperparameters were optimised via Optuna (Supplementary Table S2). Physics-prior monotonicity constraints are incorporated as an auxiliary loss term during training. Full ELBO derivation, KL annealing schedule and architecture details are in Supplementary Note A.

4.4 Forward uncertainty propagation

Forward UQ draws $N = 20{,}000$ Latin hypercube samples from the joint prior and evaluates the BNN posterior predictive mean at each sample. Using the predictive mean rather than a full predictive draw isolates parametric uncertainty from the BNN's own epistemic and aleatoric spread; the predictive-draw variant is reported in Supplementary Note D.

4.5 Sobol sensitivity analysis

Global sensitivity is assessed through variance-based Sobol indices estimated with the Saltelli cross-matrix scheme ($N_S = 4096$ per matrix, 40,960 surrogate evaluations per replication). The analysis is repeated 50 times; a factor is treated as a stable contributor only if the lower bound of its 90% CI exceeds zero. Methodological details including the Jansen estimator definition are in Supplementary Note B.

4.6 Posterior calibration

Posterior calibration uses Bayesian inference: $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})\, \pi(\boldsymbol{\theta})$, with the likelihood evaluated through the BNN posterior predictive mean. Four parameters are calibrated ($E_{\text{intercept}}$, α_{base}, α_{slope}, k_{ref}), selected on the basis of their Sobol ranking; the remaining four are fixed at their true values. The 18 benchmark cases are drawn from the test split (6 low, 6 near, 6 high stress). Observations are the true HF outputs of 5 primary coupled quantities perturbed by 2% multiplicative Gaussian noise. The posterior is sampled by Metropolis–Hastings MCMC with 4 independent chains (8000 iterations, burn-in 2000, thinning 5). Implementation details and convergence diagnostics are in Supplementary Note C.

Data availability

The coupled simulation dataset and fixed train/validation/test split used in this study are available at [repository URL upon acceptance].

Code availability

The BNN training, evaluation and analysis code are available at [repository URL upon acceptance].

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Author contributions

Y.C. developed the BNN surrogate model, designed and performed the experiments, analysed the results and drafted the manuscript. S.W. contributed to the coupled simulation framework. X.L. and T.Z. conceived and supervised the project, provided scientific guidance and revised the manuscript. All authors discussed the results and approved the final version.

Competing interests

The authors declare no competing interests.